

Sri Lanka Institute of Information Technology



Software Engineering – SE2030

Lab sheet -1

PG – 191

Project Topic – Web based Employee Management System

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Major users/stakeholders of the system

Hi, we are doing a web based employee management system project for our software engineering module. Can you help with that?



Project Discussion Bot - Year2

Thanks for sharing! I'm happy to help you with your **web-based employee management system** project for your software engineering module.

Let's first understand your project clearly. Can you please tell me:

1. Who will use this system? (For example: HR staff, employees, managers?)
2. What can users do with this system? (Like: update profiles, track performance, request training?)

Just give short answers. I'll create a company and some people for you to talk to and collect requirements.



used by HR staff, employees, managers/admin, and users can manage employee profiles, track employee attendance, generate reports, send alerts, track performance.



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Thanks! I understand now. You are building a **Web-Based Employee Management System** for your **Software Engineering module**. It will be used by **HR staff, employees, managers/admins**, and it will help with:

- Managing employee profiles
- Tracking attendance
- Generating reports
- Sending alerts
- Tracking performance

Let's imagine your client is a Sri Lankan company called **LankaTech Solutions (Pvt) Ltd**. They have around 100 employees and want to improve how they manage staff information and performance.

Here are 5 people from LankaTech who will help you gather requirements:

1. **Mr. Nuwan Perera** – HR Manager
2. **Ms. Dilani Fernando** – Admin Officer
3. **Mr. Chamika Jayasundara** – Software Engineer
4. **Ms. Tharushi Senanayake** – Team Leader
5. **Mr. Ruwan Silva** – General Manager

You can now ask questions to any of them. Just type your question, and I'll let the right person answer. 😊

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can you give another person to help gather requirements



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Great! I'll introduce a new person from LankaTech Solutions (Pvt) Ltd to help you gather more requirements.



Mr. Sanjeewa Maduranga – IT Support Officer

Hello! I help staff with technical issues and system access.

Right now, we don't have a proper way to manage user accounts. People forget passwords, and we don't know who has access to what. I want a system where I can manage user roles, reset passwords, and control access easily.

Let me know what you want to ask. 😊



Requirement Gathering

Mr. nuwan perera what problems do you have and what requirement do you need



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Hello! Thank you for your question to Mr. Nuwan Perera.



Mr. Nuwan Perera – HR Manager at LankaTech Solutions (Pvt) Ltd:

Hello. Yes, I have some problems with our current system.

We use Excel sheets and emails to manage employee data. It is slow and confusing. Sometimes we miss attendance records or forget to send alerts.

I need a system where I can easily add employee details, check attendance, and send alerts quickly. Also, I want to see reports about employee performance and attendance.

Can you help us with that?

You can now ask more questions from Mr. Nuwan, or say "next" to talk to another person. 😊

yes i can help with that is that all the problems you have or are there any

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 Mr. Nuwan Perera – HR Manager at LankaTech Solutions (Pvt) Ltd:

Thank you for asking again. Yes, there are a few more problems.

Sometimes employees forget to update their personal details. We also don't have a proper way to track promotions or training history. When someone leaves the company, it takes time to remove their records.

Also, we don't have a dashboard to see everything in one place. That would be very helpful.

Let me know if you need more details. 😊

okay thanks, can i talk to Ms. Dilani Fernando now

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 Ms. Dilani Fernando – Admin Officer at LankaTech Solutions (Pvt) Ltd:

Hello! Yes, you can talk to me.

I face problems when updating employee records. Sometimes I get wrong details or missing documents. It takes time to check everything.

I also need a way to send reminders to employees when they forget to submit forms or update their profiles. A simple dashboard would help me a lot.

Let me know what you can do for us. 😊

how do you want the system to work and what requirements do you need?

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 Ms. Dilani Fernando – Admin Officer at LankaTech Solutions (Pvt) Ltd:

Thank you for asking. I want the system to be very simple.

I should be able to log in, search for an employee, and update their details easily. I also want to send reminders with just one click. If someone forgets to update their profile or submit a form, I want the system to show me that.

Also, I want to see a small dashboard with important things like pending updates, alerts, and quick links.

Let me know if you can do that. 😊

okay thanks, can i talk to Mr. Chamika Jayasundara now

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 Hello! I'm glad you're continuing the discussion.

Let me introduce myself properly.

 Mr. Chamika Jayasundara – Software Engineer at LankaTech Solutions (Pvt) Ltd:

Hi! I'm one of the developers here. I think your system can really help us.

Right now, we don't have a proper way to connect employee data with performance tracking. Everything is manual. I want the system to be smart — like, it should show performance trends, attendance issues, and maybe even suggest training.

Also, I hope the system is fast and easy to use. If it's slow or confusing, people won't use it.

Let me know what you can do with that. 😊

what feature do you want?

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 Mr. Chamika Jayasundara – Software Engineer at LankaTech Solutions (Pvt) Ltd:

Thanks for asking. I want a few smart features.

I want a **performance dashboard** that shows how each employee is doing. It should use attendance and task data to show trends. Also, I want a **report generator** that can create monthly or yearly reports with one click.

If possible, I'd like a **notification system** that alerts us when someone's performance drops or improves.

Let me know if that's possible. 😊

    

okay thanks i want to talk to Ms. Tharushi Senanayake

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 Ms. Tharushi Senanayake – Team Leader at LankaTech Solutions (Pvt) Ltd:

Hi! Nice to meet you.

As a team leader, I want to see how my team is doing. I need to check attendance, performance, and if they are meeting deadlines. I also want to give feedback and maybe suggest training for them.

can you tell me what functional and non-functional requirements do you want

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 Ms. Tharushi Senanayake – Team Leader at LankaTech Solutions (Pvt) Ltd:

Good question. I'll keep it simple.

Functional requirements I want:

- I should be able to view my team's attendance and performance.
- I want to give feedback to my team through the system.
- I should be able to suggest training for team members.

Non-functional requirements I want:

- The system should be fast and not take too long to load.
- It should work well on both computers and phones.
- It must be easy to use, even for people who are not very technical.

okay, i would like to talk to the general manager now

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 Hello again!

Let me introduce the last person from LankaTech Solutions (Pvt) Ltd.

 Mr. Ruwan Silva – General Manager

Hello. I'm happy to support your project.

We want a system that helps us manage employees better. I need to see reports, team performance, and alerts in one place. It should help us make decisions quickly.

can you specify how you work what functional and non-functional requirements you want

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 Mr. Ruwan Silva – General Manager at LankaTech Solutions (Pvt) Ltd:

Thank you for asking. I'll explain simply.

In my work, I look at reports, team performance, and company progress. I make decisions based on that.

For **functional requirements**, I want to:

- View company-wide reports
- Get alerts about staff issues
- See team performance summaries

For **non-functional requirements**, I need:

- The system to be secure and protect employee data
- It should work fast and not crash
- It must be easy for all staff to use

hi Mr. Sanjeewa Maduranga can you tell me what are your functional and non-functional requirements

 Project Discussion Bot - Year2

 Hello again!

Let me answer as Mr. Sanjeewa Maduranga – IT Support Officer at LankaTech Solutions (Pvt) Ltd.

Hi! Thanks for asking.

For functional requirements, I want to:

- Create and manage user accounts easily.
- Reset passwords when someone forgets.
- Give access only to the right people.

For non-functional requirements, I need:

- The system to be safe and protect user data.
- It should work fast and not get stuck.
- It must be easy for staff to use without training.

Users and their Roles

1. Mr. Nuwan Perera – HR Manager

He handles employee records, attendance, and performance. He wants the system to help manage staff easily and send alerts.

2. Ms. Dilani Fernando – Admin Officer

She updates employee details and checks documents. She needs a simple dashboard and reminders for missing info.

3. Mr. Chamika Jayasundara – Software Engineer

He wants smart features like performance tracking and report generation. He also cares about speed and usability.

4. Ms. Tharushi Senanayake – Team Leader

She manages a team and gives feedback. She wants to see attendance, performance, and suggest training.

5. Mr. Ruwan Silva – General Manager

He looks at company-wide reports and makes decisions. He needs a secure, fast, and reliable system.

6. Mr. Sanjeewa Maduranga – IT Support Officer

He manages user accounts and access. He wants password reset, role control, and strong security.

Functional Requirements

1. Manage Employee Profiles.
2. Track Attendance.
3. Monitor Performance.
4. Generate Reports.
5. Send Alerts and Notifications.
6. Manage User Roles and Access.

Core Function	Description	Sub Function
Manage Employee Profiles	Add, update, and view employee details	Add new employee, Edit employee information, View employee profile, Remove employee when they leave
Track Attendance	Record and monitor daily attendance	Mark daily attendance, View attendance history, Filter by date or employee, Identify absentees
Monitor Performance	View performance trends and give feedback	Add performance feedback, View performance trends, Suggest training, Rate employee performance
Generate Reports	Create attendance and performance reports	Attendance report, Performance report, Monthly and yearly summaries, Export reports as PDF or Excel
Send Alerts and Notifications	Remind employees about updates or issues.	Send reminders for missing updates, Notify about poor attendance, Alert for upcoming

		reviews, System messages for all users
Manage User Roles and Access	Control who can access what in the system	Create user accounts, assign roles (HR, admin, employee, manager), Reset passwords, Control access to features